PROCESS FLOW DIAGRAM (PFD) — ETHANOL-WATER RECOVERY DISTILLATION TRAIN

BASIS: 700 kmol/h FEED @ 7.0 mol% ETHANOL | PRODUCT: 85.0 mol% DISTILLATE | RECOVERY: >97.5%

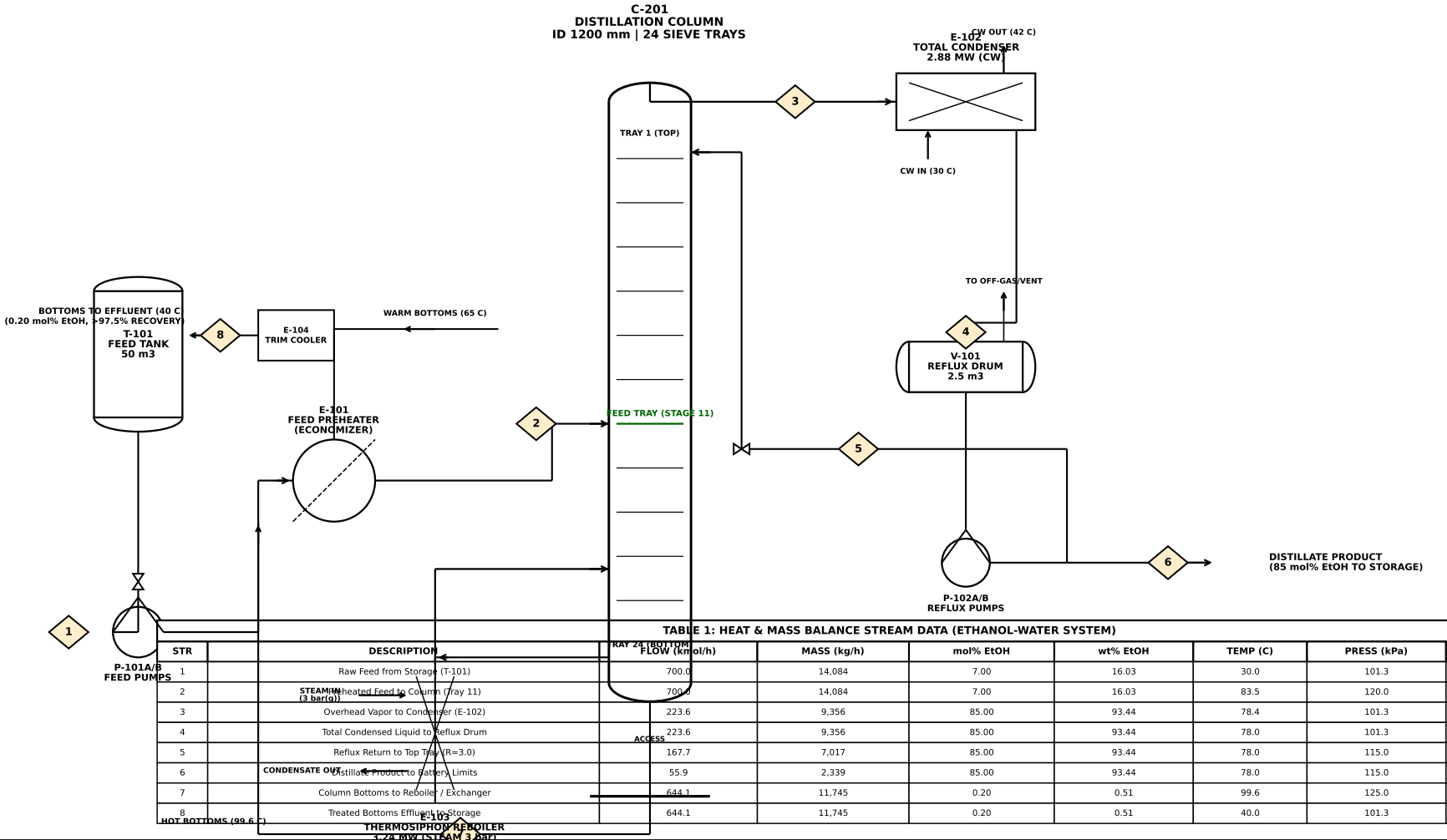

TABLE 1: HEAT & MASS BALANCE STREAM DATA (ETHANOL-WATER SYSTEM)

STR	DESCRIPTION	FLOW (kmol/h)	MASS (kg/h)	mol% EtOH	wt% EtOH	TEMP (C)	PRESS (kPa)	PHASE
1	Raw Feed from Storage (T-101)	700.0	14,084	7.00	16.03	30.0	101.3	Liquid
2	Preheated Feed to Column (Tray 11)	700.0	14,084	7.00	16.03	83.5	120.0	Bubble Liq (q=1)
3	Overhead Vapor to Condenser (E-102)	223.6	9,356	85.00	93.44	78.4	101.3	Vapor (Dew Pt)
4	Total Condensed Liquid to Reflux Drum	223.6	9,356	85.00	93.44	78.0	101.3	Subcooled Liq
5	Reflux Return to Top Tray (R=3.0)	167.7	7,017	85.00	93.44	78.0	115.0	Liquid
6	Distillate Product to Battery Limits	55.9	2,339	85.00	93.44	78.0	115.0	Liquid Product
7	Column Bottoms to Reboiler / Exchanger	644.1	11,745	0.20	0.51	99.6	125.0	Saturated Liq
8	Treated Bottoms Effluent to Storage	644.1	11,745	0.20	0.51	40.0	101.3	Liquid Effluent

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE **PROCESS FLOW DIAGRAM (PFD) & HEAT-MASS BALANCES**	LAB / SUBJECT **PROCESS EQUIPMENT DESIGN & DRAWING**	COURSE / BRANCH **B.E. CHEMICAL ENGINEERING**	SCALE **DIAGRAMMATIC**
STUDENT NAME **SOURISH SENAPATI**	ROLL NO. **002310301035**	DRAWING NO. **JU/ChE/ED/DWG-01**	DATE **01 / 09 / 2026**

ETHANOL DISTILLATION (C-201) | 700 kmol/h FEED (7 mol% EtOH) | ID 1.20 m, R=3.0 | ASME SEC VIII / TEMA